

Chapter-3

Results

This chapter discusses the data analysis and findings of the two research aims i.e., 1) explore the situations that generate the social emotions of guilt, shame and remorse, and 2) evaluate the emotional experiences of guilt, shame and remorse with the levels of appraisal. Initial data processing for each type (qualitative and quantitative) will be discussed along with the results. This allows readers to examine the thought process behind research results and minimize the chance of bias being embedded in the research process.

3.1 Phase I: Results of Qualitative Analysis

3.1.1 Aim 1

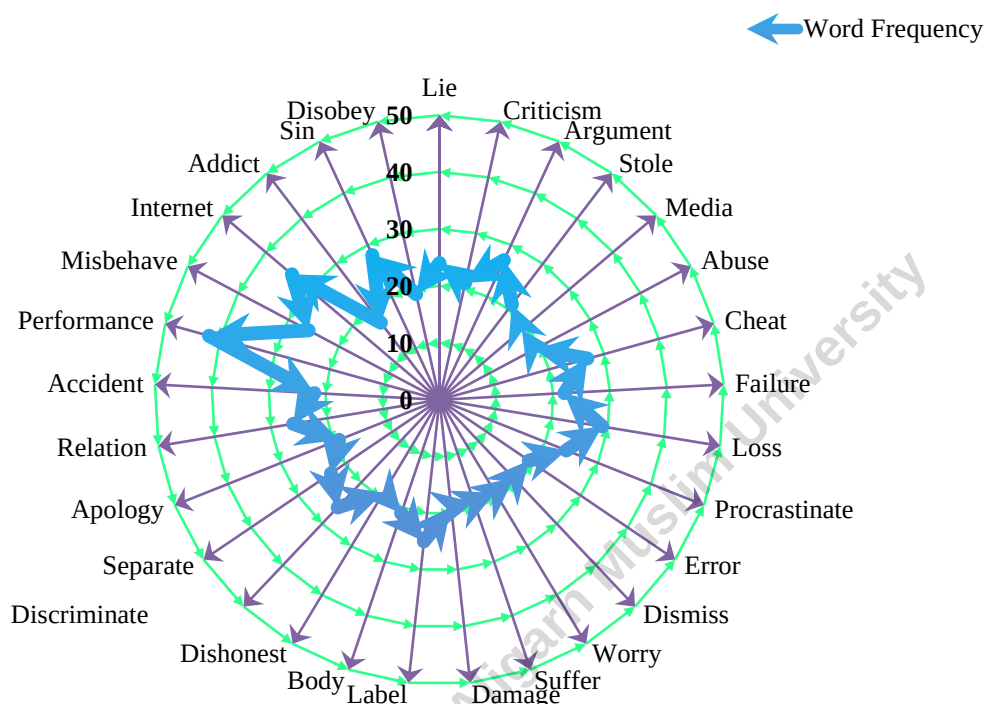
The qualitative data of each social emotion-guilt, shame and remorse was transferred one by one into the Rapid Miner 9.9 software for text mining. The goal was to use the software for generating child codes from the responses of the participants. So that it becomes easy for the readers and research community to understand the response situations of the participant's. After that from operators the processing document is applied to find out the most word occurrence (word frequency) from the responses of participants by using TF-IDF (Term Frequency-Inverse Document Frequency) (Kalra & Aggarwa, 2017). Also, the eight steps of NLP for text mining, starting form; I) Prepare the Data, II) Parts of Speech Tagging III) Tokenizing, IV) Stop Words, V) Filtering Tokens, IV) Gram (One phrase), IIV) Stemming IIIIV) and finally Codes. In this way, with the help of text mining by using NLP technique TF-IDF the 30 nodes were extracted that are most repeated in the guilt responses experienced by participants. (See Fig; 3.1)

After generating the child codes from rapid miner 9.9 version software with the help of eight steps of NLP for text mining. The codes were then matched with the participant responses one by one by using Nvivo +12 version software for the purpose of content analysis. Now, the preliminary child codes derived from the rapid miner were ready to help making parent codes (Nodes). A node is the representation of particular response and a collection of references. We can gather the references by

‘coding’ sources to a node. When we open the node we can see all the references in one place.

Figure 3.1

Grid Chart: Frequency of Most 30 Nodes Repeated in Guilt Response Situations



Therefore, the initial categorization of source data into nodes was performed and over all yield 21 parent nodes were finalized and that includes 426 references and rest of 9 child codes was dropped in content analysis due to the synonym terms and similar content was found. Therefore, the twenty one parent node categories were *Performance, Internet, Sin, Cheat, Relation, Discriminate, Misbehave, Lie, Label, Separate, Accident, Loss, Failure, Stolen, Procrastinate, Criticism, Disobey, Body, Addict, Argument and Apology*. The relative frequency of each category with references are shown in *Table 3.1*, These parent node categories were considered as situations of which the social emotions of guilt has been generated in life of individuals.

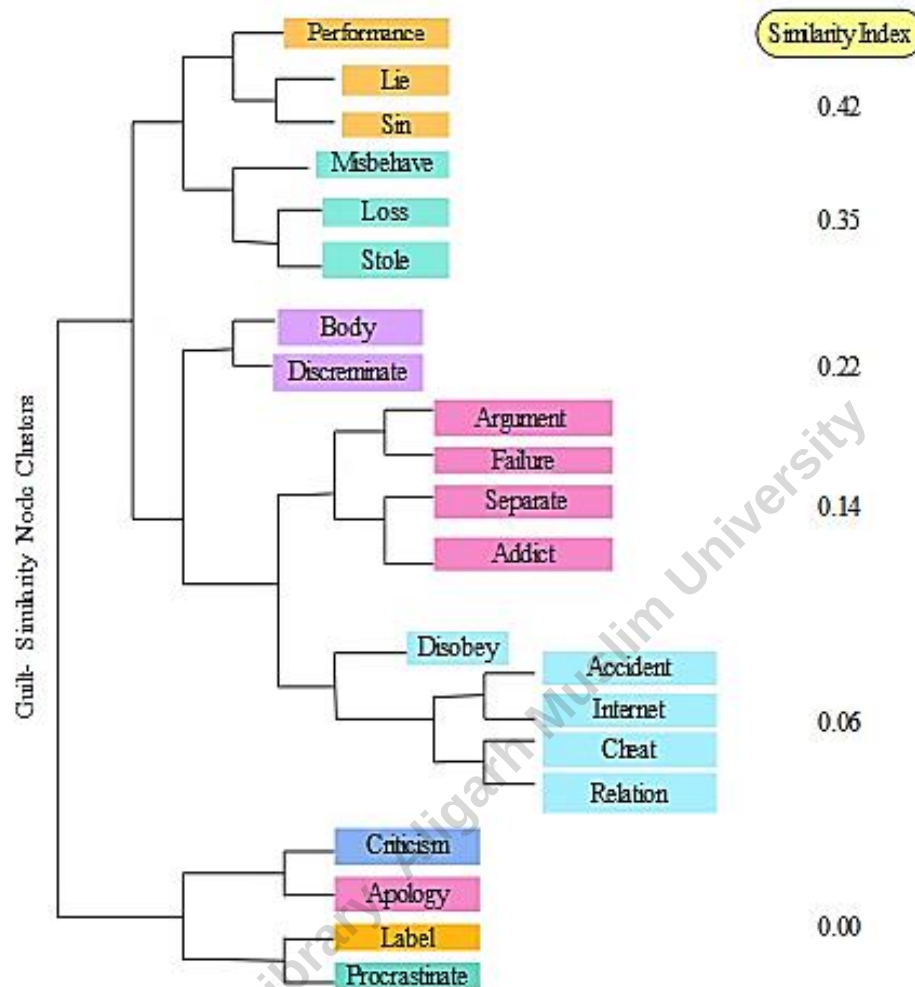
Table 3.1*Guilt Nodes with Respect to Situations and Relative Frequency Percentage*

<i>Guilt Nodes</i>	<i>Example quote</i>	<i>Relative frequency n (%)</i>
Performance	<i>"Spending money and time to coaching but not competing the examination" P-68</i>	47(11.04)
Internet	<i>"Sharing number of awful things on social media and doing hacking elements too" P-182</i>	42 (9.86)
Sin	<i>"Not having control certain prohibited things like spy and lie which are very sinful elements in my religious" P-123</i>	37(8.69)
Cheat	<i>"I feel guilt about cheating in an examination when I got caught" P-19</i>	36(8.46)
Relation	<i>"Feeling guilt by having relation with a girl before marriage which is prohibited in my religious" P-242</i>	33(7.75)
Discriminate	<i>"Feeling discrimination on the basis of my name and background on daily basis" P-156</i>	27(6.34)
Misbehave	<i>"Did bad behavior to my teacher during the lecture in the class room" P-19</i>	26(6.11)
Lie	<i>"Faking sick in order not to take a part of collage games competition" P-42</i>	24(5.64)
Label	<i>"As a Kashmiri wrong label portraits negative information towards me feel most worsen and guilt" P-189</i>	23(5.39)
Separate	<i>"Feel guilty about not being close to my family during the difficult moment occur" P-131</i>	19(4.46)
Accident	<i>"Met an accident on scotty on my mistake and got heavy damage on neck and motor damage" P-178</i>	18(4.23)
Damage	<i>"Spending money for investment which I lost due to the market collapse" P-30</i>	17(3.99)

Failure	<i>"Given bad treatment to my cousin emotionally after he fail in the examination"</i> P-91	14(3.29)
Stolen	<i>"Talking an active part with my collages by stolen a new bicycle in the campus"</i> P-12	13(3.06)
Procrastinate	<i>"Puts off studying for a test by doing unimportant chores, such as cleaning their room or baking snacks"</i> P-244	12(2.82)
Criticism	<i>"In front of elders-criticized to my sister and doing negative evaluation because of her unusual doing things"</i> P-2	12(2.82)
Disobey	<i>"Having 2000 fine while driving above the speed limit and not having proper paper work"</i> P-223	9(2.12)
Body	<i>"Due to my male voice character when speak people tease me and treat me differently and erupts my self-esteem"</i> P-60	6(1.41)
Addict	<i>"Involve and using illegal drugs such as drinking during my collage with my friends"</i> P-112	6(1.41)
Argument	<i>Guilty having argument and not to accomplish the expectations of my parents every time"</i> P-201	3(0.70)
Apology	<i>"Betraying a confidant's trust by sharing their personal information or secrets"</i> P-67	2(0.46)

Note. P= Participant

Further, the cluster analysis was carried out in Nvivo software; to see the visualize patterns of the data by grouping nodes that share similar words frequency. Therefore, the dendrogram of the cluster analysis provide graphical representation of nodes to make it easy to see similarities and differences. Nodes in the cluster analysis diagram appears close together are more similar than those that are far apart. In this study we created a cluster analysis diagram as a horizontal dendrogram as shown in fig 3.2 (a horizontal branching diagram where similar items are clustered together on the same branch and different items are further apart).

Figure 3.2*Guilt Nodes to Visualize the Word Similarity: Cluster Analysis*

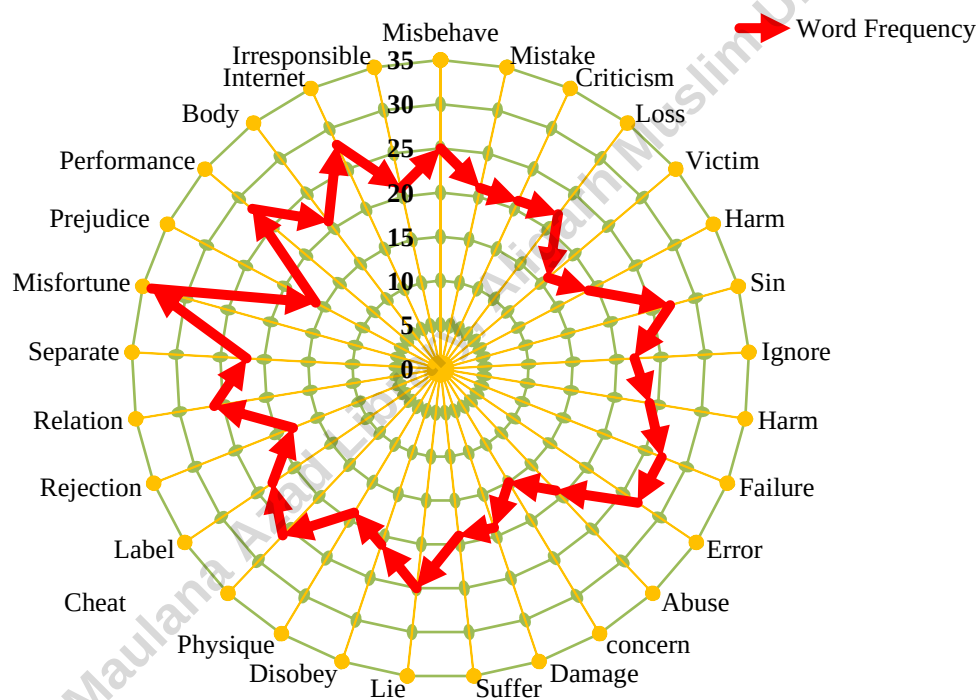
The nodes similarity in the cluster analysis is calculated with the Jaccard's coefficient. The Jaccard similarity coefficient is a statistics method used for gauging the similarity and diversity of nodes, it ranges from 0 to 1 (0=least similar, 1= most similar). Based on Jaccard's coefficient the mustard colour nodes batch (Performance, Lie and Sin) shared .42 ratio of intersection over union suggested 42% similarity, the teal colour nodes batch (Misbehave, Loss, and Stolen) shared .35 ratio of intersection over union suggested 35% similarity, the violet colour nodes batch (Body, and Discriminate) shared .22 ratio of intersection over union suggested 22% similarity, the pink colour nodes batch (Argument, Failure, Separate, and Addict) shared .14 ratio of intersection over union suggested 14% similarity, the cyan colour nodes batch (Disobey, Accident, Internet, Cheat, Relation) shared .06 ratio of intersection over union suggested 6% similarity and lastly, Criticism, Apology, Label and Procrastinate with different colour nodes batch to one another shared .00 ratio of intersection over

union suggested zero percent similarity to each other. In this way, the findings showed the 21 different situations that generate the emotions of guilt in day to day life of individuals.

For shame emotion, similarly the same pattern of coding formation was applied. Starting from Rapid Miner for word frequency and then node formation and cluster analysis generated through the NVivo +12 version software. Therefore, using NLP technique TF-IDF in rapid miner software for text mining the most 30 frequent nodes that repeated in the shame responses experienced by participant's are generated. (See fig; 3.3)

Figure 3.3

Grid Chart: Frequency of Most 30 Nodes Repeated in Shame Response Situations



The same procedures as applied in guilt emotion for generating of child codes and then identified parent nodes with the help of rapid miner and NVivo software for shame situation experiences were used. In shame response experiences, the initial categorization of source data or codes into nodes was performed and over all yield 21 parent nodes were finalized that comprises 310 references and rest of 9 child codes was dropped in content analysis due to the synonym terms and similar content was found. These twenty one parent node categories *Misfortune, Performance, Internet, Sin, Relation, Cheat, Prejudice, Misbehave, Lie, Label, Separate, Harm, Ignore,*

Loss, Failure, Mistake, Criticism, Irresponsible, Disobey, Body and Rejection. The relative frequency of each category with references are shown in Table, 3.2 in this study these parent node categories were considered as situations of which the social emotions of shame has been generated in life of individuals.

Table 3.2

Shame Nodes with Respect to Situations and Relative Frequency Percentage

Shame Nodes	Example quote	Relative frequency n (%)
Misfortune	<i>“On the railway platform, I get to slip into the water and fell off the floor. All the people around me get shocked and some of them laugh”</i> P-123	38(12.26)
Performance	<i>“On republic day, Four days before I try and prepare the poem verses. But on the day, when the anchor calls my name on the stage, due to the anxiety I forget the poem which I have to present”</i> P-64	29(9.36)
Internet	<i>“As a content writer, receives negative feedback on my work while sharing my work on social media. It feels me shameful and worthless personality”</i> P-54	28 (9.04)
Sin	<i>“Being a Muslim, I feel ashamed myself when I could not pray and do prayers on time. It feels me unethical and unreligious person every time”</i> P-178	26 (8.39)
Relation	<i>“Being shameful for having a physical relationship of another lady before getting marriage”</i> P-102	22(7.09)
Cheat	<i>“During my masters I cheat some answer in the examination hall the invigilator caught me and snatch my answer book and disrespect in front of all examinees”</i> P-154	22(7.09)
Prejudice	<i>“Feeling ashamed of the place where I grow up because of a bad name and negativity”</i> P-21	19(6.13)
Misbehave	<i>“Using an offensive language to my relatives because of certain family issues”</i> P-179	17(5.49)
Lie	<i>“Most times I constantly lie to my parents that i am going to a friend's house for college work. Rather, hanging and</i>	15(4.84)

	<i>going to parties with him”</i> P-152	
Label	<i>“Negative identity perception and people publicly criticized and humiliated me for my belongingness”</i> P-103	15(4.84)
Separate	<i>“Struggling with social isolation and feelings of loneliness”</i> P-156	13(4.19)
Harm	<i>“Involving in invading my friends privacy in front of my colleges”</i> P-25	12(3.88)
Ignore	<i>“While talking with my family about any issue the members always ignore my talk and not given any importance it feel me so embrace and shameful”</i> P-202	10(3.23)
Damage	<i>“Being unable to manage or cope with personal debts and financial losses”</i> P-23	9(2.91)
Failure	<i>“Failing to meet personal goals or resolutions”</i> P-91	8(2.59)
Mistake	<i>“Feeling ashamed of one's mistakes or shortcomings in a public setting”</i> P-72	7(2.26)
Criticism	<i>“Faced criticism and difficulty not organizing and managing project task with the batch mates”</i> P-79	7(2.26)
Irresponsible	<i>“Being unable to maintain personal boundaries or assert oneself in a relationship”</i> P-45	5(1.62)
Disobey	<i>“Feeling shame not living up to societal expectations and following the cultural norms”</i> P-76	3(0.97)
Body	<i>“I have obesity (fatty personality) and it feels shameful every time because people tease me every time and call me different bad names such as “Mota, Sandd” etc. while playing or public places”</i> P-12	3(0.97)
Rejection	<i>“Feelings of inadequacy and sensitivity to rejection for bad behaviors”</i> P-38	2(0.65)

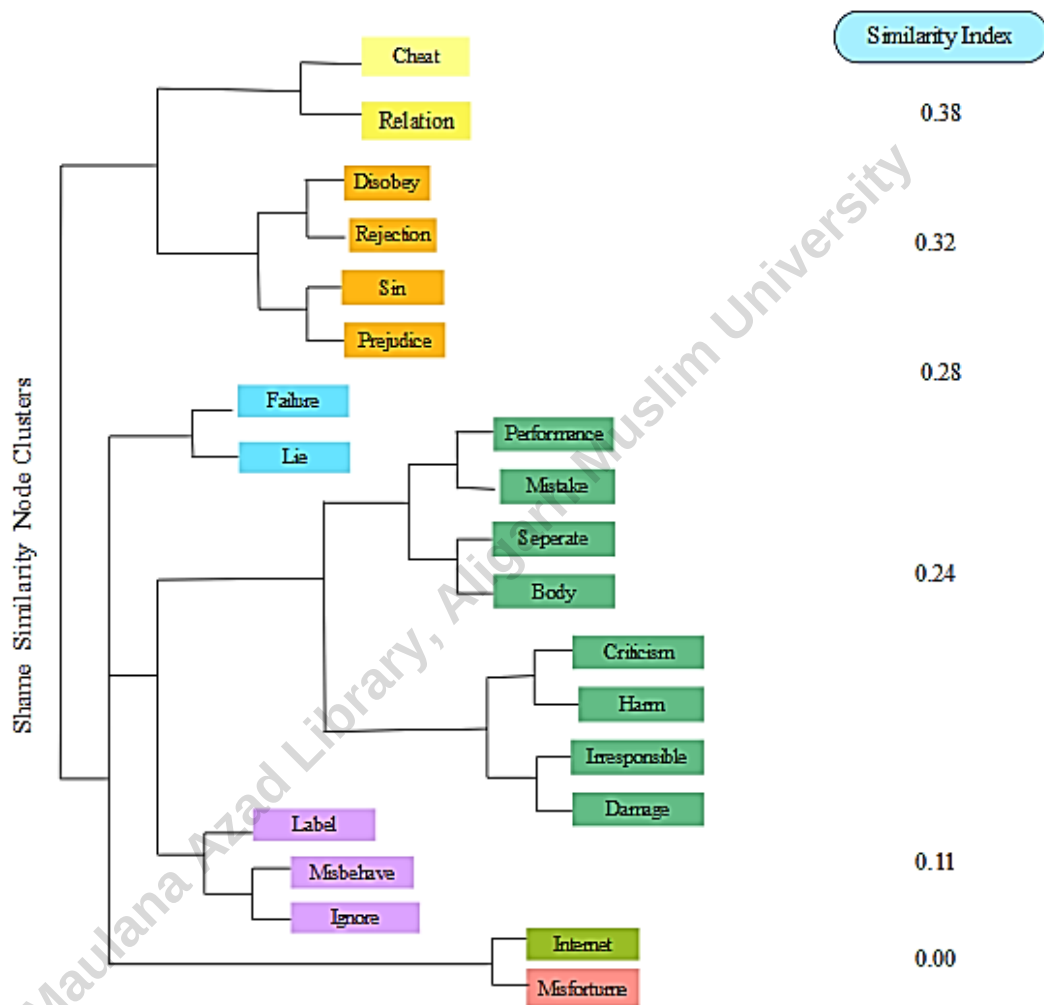
Note. P= Participant

Similarly the cluster analysis patterns for shame emotion were drawn to find out the node similarity and difference patterns. Because, the cluster analysis by dendrogram involves a hierarchical approach to grouping data points based on similarity and visualized through a tree-like diagram known as a dendrogram. Starting with individual data points, clusters are successively merged based on proximity,

forming a hierarchy. The height at which clusters merge indicates their similarity level. Dendrograms help identifying optimal cluster numbers and offer insights into hierarchical relationships within data, making them valuable for data exploration and pattern recognition. (See fig; 3.4)

Figure 3.4

Shame Nodes to Visualize the Word Similarity: Cluster Analysis



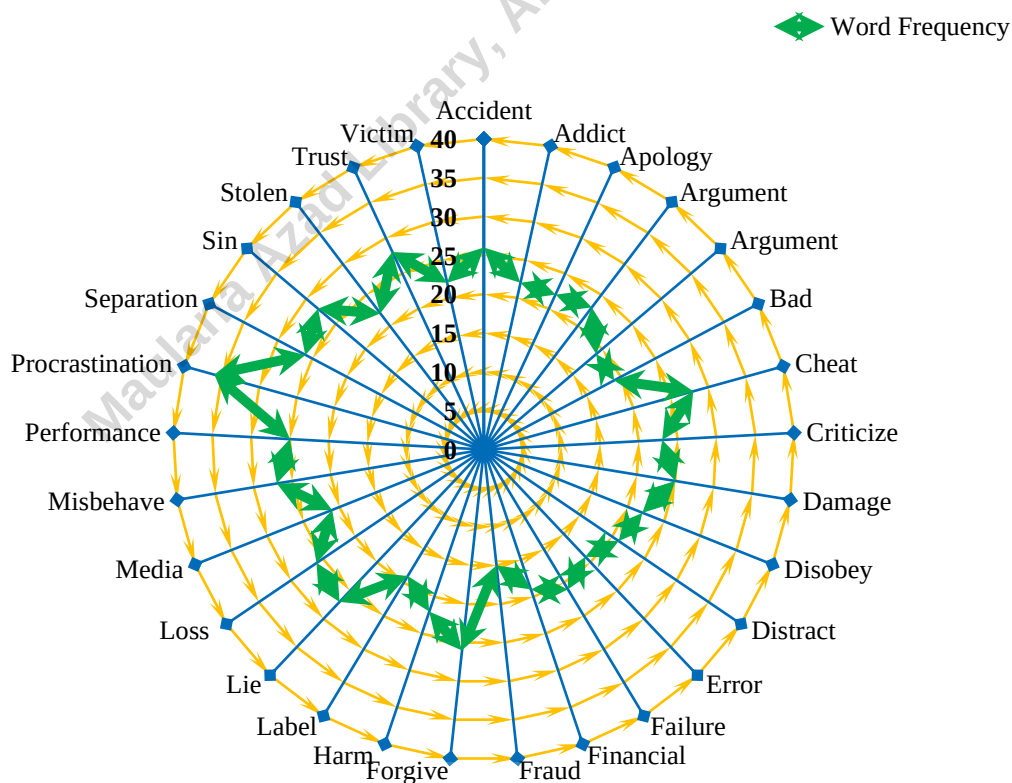
By using Jaccard's coefficient to figure out how close the nodes are in a cluster analysis and its ranges varies from 0 to 1(0= least similar, 1= most similar) as mentioned previously given in guilt cluster analysis. Based on Jaccard's coefficient the yellow colour nodes batch (Cheat and Relation) shared .38 ratio of intersection over union suggested 38% similarity, the maroon colour nodes batch (Disobey, Rejection, Sin and Prejudice) shared .32 ratio of intersection over union suggested 32% similarity, the cyan colour nodes batch (Failure and Lie) shared .28 ratio of intersection over union suggested 28% similarity, the teal colour nodes batch

(Performance, Criticism, Irresponsible, Mistake, Separate, Body, Harm and Damage) shared .24 ratio of intersection over union suggested 24% similarity, the Blue colour nodes batch (Label, Misbehave and Ignore) shared .11 ratio of intersection over union suggested 11% similarity and lastly, Internet and Misfortune with different colour nodes (Green and Bronze) batch to one another shared .00 ratio of intersection over union suggested zero percent similarity to each other. Therefore, the findings showed 21 different situations that generate the emotions of shame in day to day lives of individuals.

For remorse emotion, the same pattern of coding formation is taken place as used in guilt and shame emotion experiences. Starting with Rapid Miner for word frequency and continuing with NVivo version 12 for node formation and cluster analysis. The most 30 common terms that occur in the remorse responses reported by participants are identified by using NLP technique TF-IDF with Rapid Miner 9.9 version software for text mining. (See fig; 3.5)

Figure 3.5

Grid Chart: Frequency of Most 30 Nodes Repeated in Remorse Response Situations



The same procedure as applied in guilt and shame emotion for generating child codes and then identified parent nodes with the help of rapid miner and nvivo software for remorse situation experiences are used. In remorse response experiences, the initial categorization of source data or codes into nodes was performed and over all yield 17 parent nodes were finalized and that includes 362 references and rest of 13 child codes was dropped in content analysis due to the synonym terms and similar content was found. These seventeen parent node categories were *Procrastination, Misbehave, Addict, Argument, Apology, Performance, Lie, Damage, Disobey, Criticize, Loss, Sin, Accident, Separation, Trust, Cheat and Forgive*. The relative frequency of each category with references are shown in table 3.3 in this study these parent node categories are considered as situations of which the social emotions of remorse has been generated in life of individuals.

Table 3.3

Remorse Nodes with Respect to Situations and Relative Frequency Percentage

Remorse Nodes	Example quote	Relative frequency n (%)
Procrastination	<i>"I waste hours browsing social media and not complete and manage my homework assignment"</i> P-22	44(12.16)
Misbehave	<i>"During alive I fight with my grandparents and don't meet their standards and now when they are not with me I felt remorse every time"</i> P-221	35(9.67)
Addict	<i>"Connection with bad friend circle enhancing drugs enhancing bad smoking addiction"</i> P-67	32(8.84)
Argument	<i>"Fought with a college in the playground and exchange some vulgar and abused statement's to each other. It feels me very remorseful and want to apologize for this act"</i> P-45	29(8.02)
Apology	<i>"Want to apologize for my bad behavior with the faculty member in the university"</i> P-169	28(7.74)
Performance	<i>"Beaten to younger brother not performing well in the examination"</i> P-202	25(6.91)
Lie	<i>"Lying about having a terminal illness to gain</i>	23(6.36)

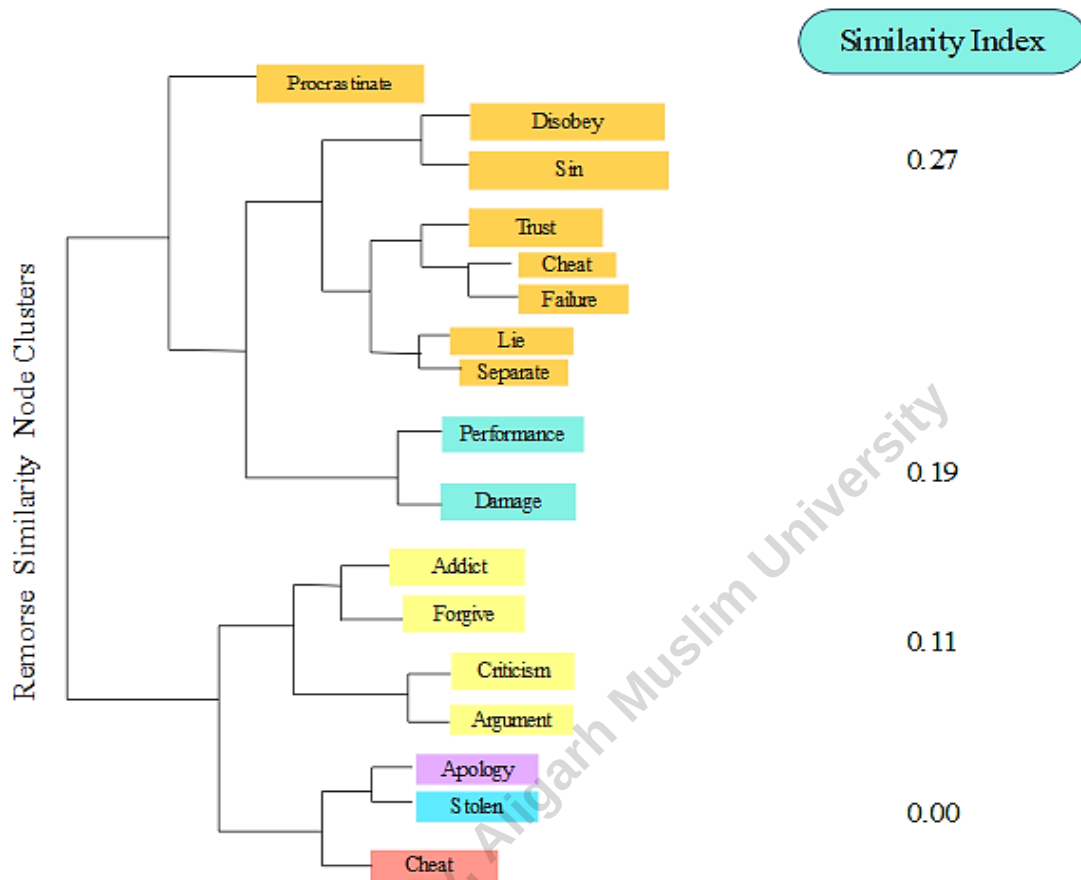
	<i>sympathy</i>	
Damage	<i>"Damaging to the collage property by broken the wall clock of the classroom"</i> _{P-17}	22(6.08)
Disobey	<i>"Lost two of my friends by disobeying safety regulations and procedures, such as failing to wear a helmet while riding a bike"</i> _{P-32}	22(6.08)
Criticize	<i>"Criticize to my friend because during covid-19, that he didn't wear the mask at the metro station, and later he fined rupees 500 for this act"</i> _{P-65}	19(5.25)
Loss	<i>"Spending money on online crypto currency that i lost because of the collapse of the market"</i> _{P-28}	19(5.25)
Sin	<i>"Every time i feel remorseful for my act that I can't stop lying, which is a very bad thing in the context of my faith"</i> _{P-89}	19(5.25)
Accident	<i>"Accidentally striking a stranger my car causes feelings of remorse and guilt"</i> _{P-56}	16(4.42)
Separate	<i>"Not living with my family for the last five years because of some personal issues"</i> _{P-12}	11(3.04)
Trust	<i>"Sharing a person's private data and secrets in a manner that breaks his trust"</i> _{P-34}	9(2.49)
Cheat	<i>"Showing falsifying documents during my interview for the post"</i> _{P-07}	5(1.39)
Forgive	<i>"Not offering any help to my brother in law who is trying to reunite and settle down the disturb family environment"</i> _{P-143}	4(1.11)

Note. P=Participant

Similarly as the cluster analysis patterns for guilt and shame emotion were drawn to find out the node similarity and difference patterns. The same method is used for the remorse emotion situation experiences. (See fig; 3.6)

Figure 3.6

Remorse Nodes to Visualize the Word Similarity: Cluster Analysis

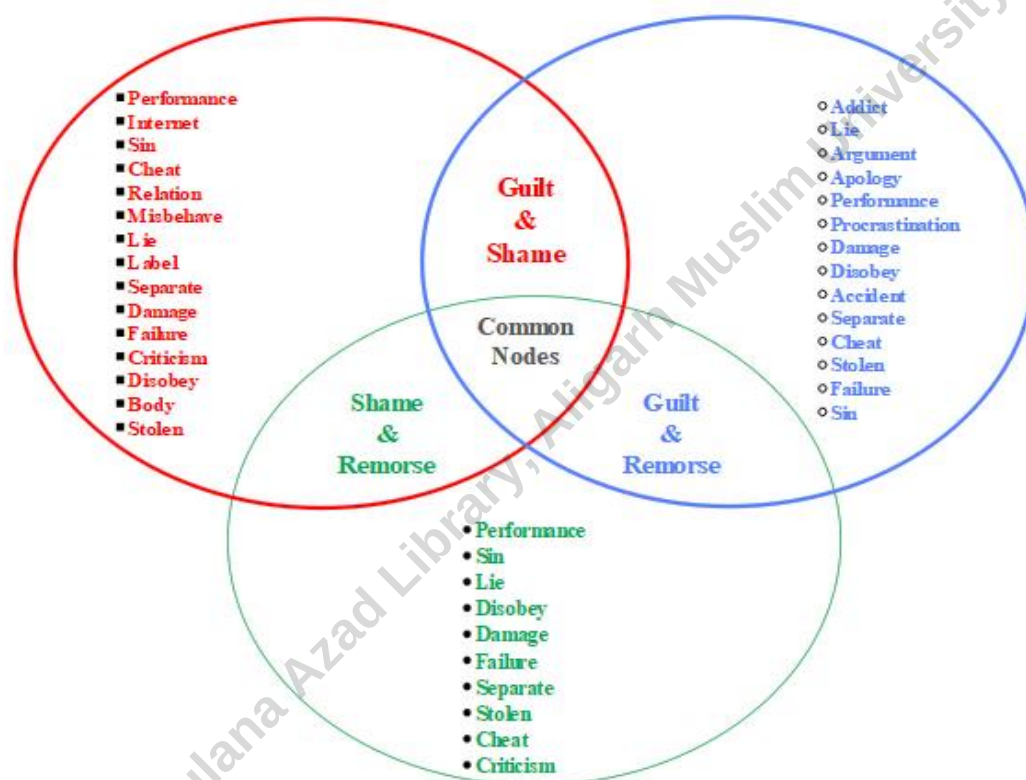


By using Jaccard's coefficient to figure out how close the nodes are in a cluster analysis and its ranges varies from 0 to 1(0=least similar, 1= most similar) as mentioned previously in guilt and shame cluster analysis. Based on Jaccard's coefficient the Coral colour nodes batch (Procrastination, Disobey, Sin, Trust, Cheat, Misbehave, Lie, Separate) shared .27 ratio of intersection over union suggested 27% similarity, the cyan nodes batch (Performance and Loss) shared .19 ratio of intersection over union suggested 19% similarity, the yellow colour nodes batch (Accident, Forgive, Damage, Criticism, and Argument) shared .11 ratio of intersection over union suggested 11% similarity, and lastly, Addict and Apology with different colour nodes (Violet and Brown) batch to one another shared .00 ratio of intersection over union suggested zero percent similarity to each other. In this way, the findings showed 17 different situations that generate the emotions of remorse in day to day lives of individuals.

Further, with the help of group query analysis technique we find the common situation nodes that are related to each other by generated the emotions of guilt, shame and remorse. Therefore, query searches to seek out elements that are related to other components of the task in a particular way as showed in Venn diagram. (See fig; 3.7)

Figure 3.7

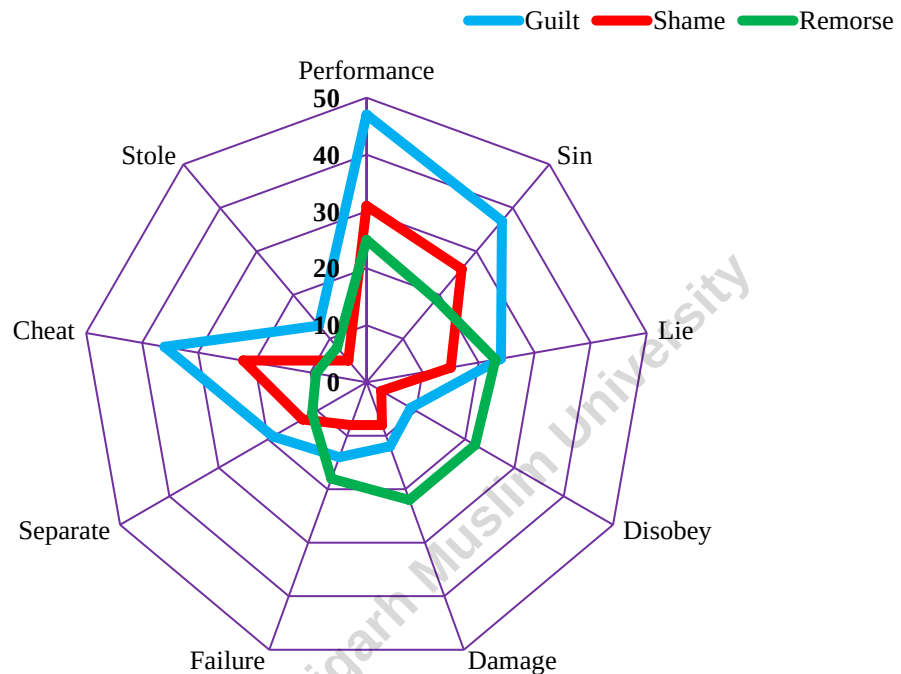
Venn Diagram: Common Nodes Altogether Sharing by Guilt and Shame, Shame and Remorse, Guilt and Remorse Emotional Experience



The results show that there were 15 situations common for evocation of guilt and shame, 14 situations specifically trigger the emotions of guilt and remorse, and finally 10 situations altogether elicit the emotions of shame and remorse. In this way, these findings highlight the situations altogether of guilt, shame and remorse and provide evidences for understanding these social emotions in various situations.

Figure 3.8

Group Query Analysis: Common Nodes Altogether Sharing by Guilt, Shame and Remorse Emotional Situations



The results shows that with the help of group query the nine situations were altogether commonly generated the emotions of guilt, shame and remorse. However, there was a variance in frequency of these nine common situations. *The blue line* shows the frequency of guilt emotions in which the frequency of performance, sin, lie, cheat, was high as compared to stolen, separate, failure and damage. Similarly, the *red line* shows the frequency of shame common situations in which the frequency of performance, sin, lie, damage and cheat was high as compared to the frequency of disobeys and stole. And finally, the *green line* shows the frequency of remorse situations in which the frequency of performance, lie, disobey, damage and failure is high as compared to the situation of separate, cheat, and stolen. In this way, the situations was identified separately and altogether during the emotions of guilt, shame and remorse and fulfill our aim 1 of research i.e.; explore the situations (separately and altogether) generate the social emotions of guilt, shame and remorse.

3.2 Phase II: Results of Quantitative Analysis

3.2.1 Aim 2

Before applying the parametric statistical tests the data screening procedure was carried out to ensure that the data must be spotless and fulfilling the mandatory assumptions of the parametric statistical tests that are used in this research. Further, the results of the study are reported by using APA 7th edition publication manual (Bradley et al., 2020).

Table 3.4

Chi-Square Test of Independence: Subjective Feeling State of Guilt by Levels of Appraisal (N=256)

Guilt Emotion	Low		Moderate		High		$\chi^2(4)$	V
Subjective State	Appraisal		Appraisal		Appraisal			
	N	%	N	%	N	%		
Guilt Occur								
6 Months Ago	57	31.1	60	34.9	55	32	29.97***	.35
12 Months Ago	7	17.1	25	61.0	9	22		
More than Year	15	34.9	3	7.0	25	58.1		
Guilt Duration								
Days	42	48.3	42	48.3	3	3.4	58.34***	.47
Months	16	18.8	25	29.4	44	51.8		
Years	21	25.0	21	25.0	42	50.0		
Guilt Reoccurrence								
Few Days	6	15.4	16	41.0	17	43.6	9.86**	.19
Few Months	28	26.7	39	37.1	38	36.2		
Few Years	45	40.2	33	29.5	34	30.4		
Guilt Intensity								
Somewhat Intensive	34	47.2	34	47.2	4	5.6	39.04***	.39
Moderately Intensive	24	23.5	33	32.4	45	44.1		
Very High Intensive	21	25.6	21	25.6	40	48.8		

*** $p < .001$, ** $p < .01$

Table 3.4 shows the chi-square test to examine the independence among levels of appraisal and guilt subjective feeling state. The result displays the significant independence among the levels of appraisal and the four (occurrence, duration, reoccurrence and intensity) domains of subject feeling state of guilt.

For *guilt occurrence*, the result showed significant independence among levels of appraisal and guilt occurrence levels with $\chi^2 (4, N=256) = 29.97, p < .001, V=.35$. The value of Cramer's V was .35 which indicated the large effect size. The findings show that adult whose guilt occurs before the first six and twelve months ago displayed high frequency of moderate appraisal level as compared to the low and high. Therefore, the results revealed that guilt occurrence after one year exhibited high frequency at high level of appraisal then low and medium appraisal levels.

Similarly, the results of *guilt duration* showed the significant independence among levels of appraisal and guilt duration state levels with $\chi^2 (4, N=256) = 58.34, p < .001, V=.47$. The value of Cramer's V was .47 which indicated the large effect size. The findings show that adult whose guilt duration lasts in days showing more frequency at low and medium appraisal level then high. However, the guilt duration that lasts in months and years showing more frequency at medium and high appraisal level.

Further, for *guilt reoccurrence subject state* the result showed significant independence among levels of appraisal and guilt reoccurrence with $\chi^2 (4, N=256) = 9.86, p < .01, V=.19$. The value of Cramer's V was .19 which indicated Medium effect size. The findings show that adult whose guilt reoccurrence within early few days and months displayed high frequency of guilt reoccurrence at moderate and high appraisal level as compared to the low level. Further, the findings shows that guilt reoccurrence after few year exhibited high frequency at low level of appraisal then medium and high appraisal levels.

Lastly, for *guilt intensity* subject state showed significant independence among the appraisal levels and guilt intensity state with $\chi^2 (4, N=256) = 39.04, p < .001, V=.39$. The value of Cramer's V was .39 which indicated large effect size. The findings show that adult whose feel somewhat guilt intensity lasts in the early days showing more frequency at low and medium appraisal level then high. Also, the moderately and very high intensive guilt situations showing more frequency at

medium and high appraisal level. Hence, the guilt distribution with level of appraisal followed by the hypothesis H_1 that “there will be independence in cell frequencies for guilt subjective feelings of state by levels of appraisal” was supported in the present study.

Table 3.5

Guilt Emotion Experiences With Levels of Appraisal: One Way MANOVA (N=256)

Guilt Emotion Experiences	Low Appraisal		Moderate Appraisal		High Appraisal		F(2)	η^2	Post-Hoc Interp.
	M	SD	M	SD	M	SD			
Bodily Symptoms	3.56 _a	1.08	6.15 _b	0.89	8.07 _c	0.98	234.74***	.65	a<b<c
Expressive Reactions	2.59 _a	1.24	5.42 _b	1.06	7.49 _c	0.99	469.23***	.79	a<b<c
Verbal Reactions	2.32 _a	1.06	3.61 _b	1.05	5.69 _c	1.02	421.08***	.77	a<b<c
Psychological Consequences	3.09 _a	1.12	4.78 _b	0.98	6.65 _c	1.09	224.69***	.65	a<b<c

*** $p < .001$.

Table 3.5 shows mean, standard deviation and F-value for four domains of emotional experiences (bodily symptoms, expressive reaction, verbal reactions and psychological consequences) with the levels of appraisal. Results indicate significant mean differences across the levels of appraisal on *bodily symptoms* with $F(2, 256) = 234.74$, $p < .001$. Findings revealed that high level of appraisal exhibited high level of bodily symptoms as compared to moderate and low appraisal levels. The value of η^2 was .65 which indicates large effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Further, Results indicate significant mean differences across the levels of appraisal on *expressive reactions* with $F(2, 256) = 469.23$, $p < .001$. Findings revealed that high level of appraisal exhibited high level of expressive reactions as compared to moderate and low appraisal levels. The value of η^2 was .79 which indicates large effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Similarly, the results indicate significant mean differences across the levels of appraisal on *verbal reactions* with $F(2, 256) = 421.08$, $p < .001$. Findings revealed that high level of appraisal exhibited high level of verbal reactions as compared to moderate and low appraisal levels. The value of η^2 was .77 which indicates large

effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Lastly, the results indicate significant mean differences across the levels of appraisal on *psychological consequences* with $F(2, 256) = 224.69, p < .001$. Findings reveled that high level of appraisal exhibited high level of psychological consequences as compared to moderate and low appraisal levels. The value of η^2 was .65 which indicates large effect size. Based on these findings the hypothesis H_2 “*there will be difference among guilt emotion experiences with the levels of appraisal*” was supported. The Post-Hoc (Tuckey) comparisons also indicated significant mean difference with each levels of appraisal.

Table 3.6

Chi-Square Test of Independence: Subjective Feeling State of Shame by Levels of Appraisal (N=256)

Shame Emotion Subjective State	Low		Moderate		High		$\chi^2(4)$	V
	Appraisal		Appraisal		Appraisal			
	N	%	N	%	N	%		
Shame Occur								
6 Months Ago	29	30.9	28	29.8	37	39.4	18.94**	.19
12 Months Ago	29	46.0	11	17.5	23	36.5		
More	17	17.2	24	24.2	58	58.6		
Shame Duration								
Days	38	100	0	0.0	0	0.0	202.79****	.62
Months	26	20.6	63	50.0	37	29.4		
Years	11	12.0	0	0.0	81	88.0		
Shame Reoccurrence								
Few Days	21	95.5	1	4.5	0	0.0	186.32****	.60
Few Months	10	9.4	0	0.0	96	90.6		
Few Years	44	34.4	62	48.4	22	17.2		
Shame Intensity								
Somewhat Intensive	27	34.6	25	32.1	26	33.3	21.36***	.20
Moderately Intensive	31	41.9	13	17.6	30	40.5		
Very High Intensive	17	16.3	25	24.6	62	59.6		

** $p < .01$, *** $p < .001$.

Table 3.6 performed the chi-square test to examine the independence among levels of appraisal and shame subjective feeling state. The result shows the significant independence among levels of appraisal and four (occurrence, duration, reoccurrence and intensity) domains of subject feeling state of shame.

For *shame occurrence*, the result showed significant independence among levels of appraisal and guilt occurrence levels with $\chi^2 (4, N=256) = 18.94, p < .01, V = .19$. The value of Cramer's V was .19 which indicated the medium effect size. The findings show that adult whose shame occurs within the first six and twelve months ago displayed high frequency of moderate appraisal level as compared to the low and high. Further, the findings revealed that shame occurrence after one year exhibited high frequency at high level of appraisal then low and medium appraisal levels.

Similarly, the *shame duration* the result showed significant independence among levels of appraisal and shame duration state levels with $\chi^2 (4, N=256) = 202.79, p < .001, V = .62$. The value of Cramer's V was .62 which indicated the large effect size. The findings show that adult whose shame duration lasts in days showing more frequency at low and medium appraisal level then high. However, the shame duration that lasts in months and years showing more frequency at medium and high appraisal level.

Further, for *shame reoccurrence state* result showed significant independence among levels of appraisal and shame reoccurrence with $\chi^2 (4, N=256) = 186.32, p < .001, V = .60$. The value of Cramer's V was .60 which indicated large effect size. The findings show that adult whose shame reoccurrence within early few days and months displayed high frequency of shame reoccurrence at moderate and high appraisal level as compared to the low level. Further, the findings shows that shame reoccurrence after few year exhibited high frequency at low level of appraisal then medium and high appraisal levels.

Lastly, for *shame intensity state* the result showed significant independence among the appraisal levels and shame intensity state with $\chi^2 (4, N=256) = 21.36, p < .001, V = .20$. The value of Cramer's V was .20 which indicated medium effect size. The findings show that adult whose feel somewhat shame intensity lasts in the early days showing more frequency at low and medium appraisal level then high. Also, the moderately and very high intensive shame situations showed more frequency at medium and high appraisal level. Hence, the shame distribution with level of appraisal followed by the hypothesis H_3 that "*there will be independence in cell frequencies for shame subjective feelings of state by levels of appraisal*" was supported in the present study.

Table 3.7*Shame Emotion Experiences With Levels of Appraisal: One Way MANOVA (N=256)*

Shame Emotion Experiences	Low Appraisal		Moderate Appraisal		High Appraisal		<i>F</i> (2)	η^2	Post-Hoc Interp.
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			
Bodily Symptoms	2.60 _a	0.98	4.83 _b	0.75	7.58 _c	1.23	468.39***	.79	a<b<c
Expressive Reactions	2.75 _a	1.15	4.49 _b	1.04	6.94 _c	1.25	523.79***	.81	a<b<c
Verbal Reactions	2.29 _a	1.02	4.65 _b	1.01	6.65 _c	0.97	307.86***	.70	a<b<c
Psychological Consequences	2.40 _a	1.55	5.97 _b	1.11	8.23 _c	1.21	339.19***	.73	a<b<c

*** $p < .001$.

Table 3.7 shows mean, standard deviation and *F*-value for bodily symptoms, expressive reaction, verbal reactions and psychological consequences across the levels of appraisal. Results indicate significant mean differences across the levels of appraisal on *bodily symptoms* with $F(2, 256) = 468.39, p < .001$. Findings revealed that high level of appraisal exhibited high level of bodily symptoms as compared to moderate and low appraisal levels. The value of η^2 was .79 which indicates high effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Further, Results indicate significant mean differences across the levels of appraisal on *expressive reactions* with $F(2, 256) = 523.79, p < .001$. Findings revealed that high level of appraisal exhibited high level of expressive reactions as compared to moderate and low appraisal levels. The value of η^2 was .81 which indicates moderate effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Similarly, the results indicate significant mean differences across the levels of appraisal on *verbal reactions* with $F(2, 256) = 307.86, p < .001$. Findings revealed that high level of appraisal exhibited high level of verbal reactions as compared to moderate and low appraisal levels. The value of η^2 was .70 which indicates moderate effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Lastly, the results indicate significant mean differences across the levels of appraisal on *psychological consequences* with $F(2, 256) = 339.19, p < .001$. Findings

reveled that high level of appraisal exhibited high level of psychological consequences as compared to moderate and low appraisal levels. The value of η^2 was .65 which indicates moderate effect size. Based on these findings the hypothesis H₄ “there will be difference among shame emotion experiences with the levels of appraisal” was supported. The Post-Hoc (Tuckey) comparisons also indicated significant mean difference with each levels of appraisal.

Table 3.8

Chi-Square Test of Independence: Subjective Feeling State of Remorse by Levels of Appraisal (N=256)

Remorse Emotion Subjective State	Low		Moderate		High		$\chi^2(4)$	V
	Appraisal		Appraisal		Appraisal			
	N	%	N	%	N	%		
Remorse Occur								
6 Months Ago	81	46.8	40	23.1	52	30.1	44.89***	.41
12 Months Ago	13	32.5	9	22.5	18	45.0		
More	2	4.7	30	69.8	11	25.6		
Remorse Duration								
Days	54	67.5	9	11.3	17	21.3	63.59***	.49
Months	42	31.3	44	32.8	48	35.8		
Years	0	0.0	26	61.9	16	38.1		
Remorse Reoccurrence								
Few Days	7	11.9	21	35.6	31	52.5	25.84***	.32
Few Months	50	43.1	33	28.4	33	28.4		
Few Years	39	48.1	25	30.9	17	21.0		
Remorse Intensity								
Somewhat Intensive	44	63.8	8	11.6	17	24.6	35.62***	.38
Moderately Intensive	38	30.2	41	32.5	47	37.3		
Very High Intensive	14	23.0	30	49.2	17	27.9		

*** $p < .001$.

Table 3.8 performed the chi-square test to examine the independence among levels of appraisal and remorse subjective feeling state. The result shows the significant independence among the levels of appraisal and four (occurrence, duration, reoccurrence and intensity) domains of subject feeling state of remorse.

For remorse occurrence, the result showed significant independence among levels of appraisal and remorse occurrence levels with $\chi^2(4, N=256) = 44.89$, $p < .001$, $V = .41$. The value of Cramer's V was .41 which indicated the large effect size. The findings show that adult whose remorse occurs within the first six and twelve

months ago displayed high frequency of moderate appraisal level as compared to the low and high. Further, the findings revealed that remorse occurrence after one year exhibited high frequency at high level of appraisal then low and medium appraisal levels.

Similarly, the *remorse duration* the result showed independence among levels of appraisal and remorse duration state levels with $\chi^2 (4, N=256) = 63.59, p < .001, V=.49$. The value of Cramer's V was .49 which indicated the large effect size. The findings show that adult whose remorse duration lasts in days showing more frequency at low and medium appraisal level then high. However, the remorse duration that lasts in months and years showing more frequency at medium and high appraisal level.

Further, for *remorse reoccurrence state* result showed significant independence among levels of appraisal and remorse reoccurrence with $\chi^2 (4, N=256) = 25.84, p < .001, V=.32$. The value of Cramer's V was .32 which indicated medium effect size. The findings show that adult whose remorse reoccurrence within early few days and months displayed high frequency of remorse reoccurrence at moderate and high appraisal level as compared to the low level. Further, the findings shows that remorse reoccurrence after few year exhibited high frequency at low level of appraisal then medium and high appraisal levels.

Lastly, for *remorse intensity* state the result showed significant independence among the appraisal levels and remorse intensity state with $\chi^2 (4, N=256) = 35.62, p < .001, V= .38$. The value of Cramer's V was .38 which indicated large effect size. The findings show that adult whose feel somewhat remorse intensity lasts in the early days showing more frequency at low and medium appraisal level then high. Further the moderate and high intensive remorse situations showing more frequency at medium and high appraisal levels. Hence, the remorse distribution with level of appraisal followed by the hypothesis H₅ that "*there will be independence in cell frequencies for remorse subjective feelings of state by levels of appraisal*" was supported in the present study.

Table 3.9

Remorse Emotion Experiences With Levels of Appraisal: One Way MANOVA (N=256)

Remorse Emotion Experiences	Low Appraisal		Moderate Appraisal		High Appraisal		F(2)	η^2	Post-Hoc Interp.
	M	SD	M	SD	M	SD			
Bodily Symptoms	4.02 _a	1.01	6.10 _b	0.89	8.09 _c	0.68	356.38***	.75	a<b<c
Expressive Reactions	3.67 _a	1.33	5.29 _b	1.09	8.01 _c	0.93	467.35***	.79	a<b<c
Verbal Reactions	2.08 _a	0.87	3.24 _b	1.02	5.91 _c	1.06	322.16***	.71	a<b<c
Psychological Consequences	2.19 _a	0.87	3.58 _b	0.71	5.30 _c	0.72	349.69***	.73	a<b<c

*** $p < .001$

Table 3.9 shows mean standard deviation and F-value for bodily symptoms, expressive reaction, verbal reactions and psychological consequences across the levels of appraisal. Results indicate significant mean differences across the levels of appraisal on *bodily symptoms* with $F(2, 256) = 356.38, p < .001$. Findings revealed that high level of appraisal exhibited high level of bodily symptoms as compared to moderate and low appraisal levels. The value of η^2 was .75 which indicates large effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Further, Results indicate significant mean differences across the levels of appraisal on *expressive reactions* with $F(2, 256) = 467.35, p < .001$. Findings revealed that high level of appraisal exhibited high level of expressive reactions as compared to moderate and low appraisal levels. The value of η^2 was .79 which indicates large effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Similarly, the results indicate significant mean differences across the levels of appraisal on *verbal reactions* with $F(2, 256) = 322.16, p < .001$. Findings revealed that high level of appraisal exhibited high level of expressive reactions as compared to moderate and low appraisal levels. The value of η^2 was .71 which indicates large effect size. The Post-Hoc (Tuckey) comparisons indicated significant mean difference with each levels of appraisal.

Lastly, the results indicate significant mean differences across the levels of appraisal on *psychological consequences* with $F(2, 256) = 234.74, p < .001$. Findings revealed that high level of appraisal exhibited high level of psychological consequences as compared to moderate and low appraisal levels. The value of η^2 was .65 which indicates large effect size. Based on these findings the hypothesis H_6 “*there will be difference among remorse emotion experiences with the levels of appraisal*” was supported. The Post-Hoc (Tuckey) comparisons also indicated significant mean difference with each levels of appraisal.

Table 3.10

Comparison of Guilt, Shame and Remorse Experiences by Participants Residential Status (N=256)

GSR Experiences	J&K		UP		<i>t</i> (254)	Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Guilt Experiences						
G-Physiological Symptoms	6.55	2.43	5.58	1.59	5.99***	0.26
G-Expressive Reactions	5.90	2.54	4.76	1.89	3.83***	0.36
G-Verbal Reactions	4.76	1.78	3.26	1.37	4.12***	0.94
G-Psychological Consequences	5.61	1.97	4.34	1.42	7.59***	0.73
Shame Experiences						
S-Physiological Symptoms	5.59	2.28	5.32	2.46	.908	0.11
S-Expressive Reactions	5.59	2.38	5.32	1.93	1.79	0.11
S-Verbal Reactions	5.16	1.95	5.04	1.84	.508	0.06
S-Psychological Consequences	6.00	3.03	5.94	2.58	.182	0.02
Remorse Experiences						
R-Physiological Symptoms	5.93	1.56	5.96	2.14	.142	0.03
R-Expressive Reactions	5.38	1.94	5.67	2.27	1.08	0.13
R-Verbal Reactions	3.39	1.38	3.79	2.20	1.28	0.21
R-Psychological Consequences	3.49	1.47	3.70	1.54	1.12	0.14

*** $p < .001$

Note. $n=115$ (J&K) and $n=141$ (UP)

G= Guilt, S= Shame, R=Remorse, J&K=Jammu & Kashmir and UP= Uttar-Pradesh

Table 3.10 revealed significant mean differences on guilt experiences of physiological symptoms with $t(254) = 5.99, p < .001$. Findings showed that the Jammu and Kashmir adults exhibited higher scores on physiological symptoms ($M=6.55, SD=2.43$) as compared to the Uttar-Pradesh adults ($M=5.58, SD=1.59$).

The value of Cohen's d was .26 which indicates the small effect size. Further, there is a significant mean differences on guilt experiences of expressive reactions with $t(254) = 3.83, p < .001$. Findings showed that the Jammu and Kashmir adults exhibited higher scores on expressive reactions ($M=5.905, SD=2.54$) as compared to the Uttar-Pradesh adults ($M=4.76, SD=1.89$). The value of Cohen's d was .36 which indicates the medium effect size. Correspondingly, there is a significant mean differences on guilt experiences of verbal reactions with $t(254) = 4.12, p < .001$. Findings showed that the Jammu and Kashmir adults exhibited higher scores on expressive reactions ($M=4.76, SD=1.78$) as compared to the Uttar-Pradesh adults ($M=3.26, SD=1.37$). The value of Cohen's d was .94 which indicates the large effect size. Lastly, there is a significant mean differences on guilt experiences of psychological consequences with $t(254) = 7.59, p < .001$. Findings showed that the Jammu and Kashmir adults exhibited higher scores on psychological consequences ($M=5.61, SD=1.97$) as compared to the Uttar-Pradesh adults ($M=4.34, SD=1.42$). The value of Cohen's d was .73 which indicates the large effect size.

However, findings indicating non-significant mean differences of *shame experiences* of bodily symptoms with $t(254) = .908, p = .365$, expressive reactions with $t(254) = 1.97, p = .076$, verbal reactions with $t(254) = .508, p = .612$ and psychological consequences with $t(254) = .182, p = .856$. The value of Cohen's d was .11, .11, .06 and .02 which means small effect size's in all the four experiences.

Similarly, findings indicating non-significant mean differences of *remorse experiences* of bodily symptoms with $t(254) = .142, p = .887$, expressive reactions with $t(254) = 1.08, p = .279$, verbal reactions with $t(254) = 1.28, p = .205$ and psychological consequences with $t(254) = 1.12, p = .271$. The value of Cohen's d was .03, .13, .21 and .14 which means small effect size's in all four experiences. Hence, the hypothesis H_7 "there will be the difference in guilt, shame and remorse experiences between the participants of Jammu & Kashmir and Uttar Pradesh" was supported in guilt social emotion experiences but not in the context of shame and remorse.

Table 3.11

Comparison of Guilt, Shame and Remorse Experiences by Participants Gender (N=256)

GSR Experiences	Male		Female		<i>t</i> (254)	Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Guilt Experiences						
G-Physiological Symptoms	6.05	2.25	5.99	1.93	.209	0.03
G-Expressive Reactions	5.11	2.52	5.38	2.07	.942	0.12
G-Verbal Reactions	4.14	1.68	3.80	1.75	1.53	0.19
G-Psychological Consequences	4.97	1.91	4.87	1.72	.444	0.06
Shame Experiences						
S-Physiological Symptoms	6.78	2.07	4.54	2.15	8.26***	1.06
S-Expressive Reactions	6.50	1.98	4.17	1.71	10.03***	1.25
S-Verbal Reactions	5.84	1.76	4.58	1.80	5.53***	0.71
S-Psychological Consequences	7.67	2.40	4.82	2.41	9.28***	1.18
Remorse Experiences						
R-Physiological Symptoms	5.28	1.89	6.40	1.17	4.80***	0.72
R-Expressive Reactions	4.75	1.69	6.08	2.13	5.12***	0.69
R-Verbal Reactions	3.05	1.89	4.06	2.13	4.36***	0.51
R-Psychological Consequences	3.28	1.31	2.82	1.58	2.83***	0.68

*** $p < .001$

Note. $n = 103$ (Male) and $n = 153$ (Female)

G= Guilt, S= Shame, R=Remorse

Table 3.11 shows the non-significant mean differences of *guilt experiences* of bodily symptoms with $t(254) = .209$, $p = .834$, expressive reactions with $t(254) = .942$, $p = .347$, verbal reactions with $t(254) = 1.53$, $p = .125$ and psychological consequences with $t(254) = .444$, $p = .658$. The value of Cohen's d was .03, .12, .19 and .06 which means small effect size's in all four experiences.

However, significant mean differences on shame experiences of physiological symptoms with $t(254) = 8.26$, $p = .001$. Findings showed that the male adults exhibited higher scores on physiological symptoms ($M = 6.78$, $SD = 2.15$) as compared to the female adults ($M = 4.54$, $SD = 2.15$). The value of Cohen's d was 1.06 which indicates the large effect size. Further, there is a significant mean differences on shame experiences of expressive reactions with $t(254) = 10.03$, $p < .001$. Findings showed that the male adults exhibited higher scores on expressive reactions ($M = 6.50$,

$SD=1.98$) as compared to the female adults ($M=4.17$, $SD=1.71$). The value of Cohen's d was 1.25 which indicates the large effect size. Correspondingly, there is a significant mean differences on shame experiences of verbal reactions with $t(254) = 5.53$, $p < .001$. Findings showed that the male adults exhibited higher scores on expressive reactions ($M=5.84$, $SD=1.76$) as compared to the female adults ($M=4.58$, $SD=1.80$). The value of Cohen's d was 0.71 which indicates the large effect size. Lastly, there is a significant mean differences on shame experiences of psychological consequences with $t(254) = 9.28$, $p < .001$. Findings showed that the male adults exhibited higher scores on psychological consequences ($M=7.67$, $SD=2.40$) as compared to the female adults ($M=4.82$, $SD=2.41$). The value of Cohen's d was 1.18 which indicates the large effect size.

Similarly, significant mean differences on remorse experiences of physiological symptoms with $t(254) = 4.80$, $p < .001$. Findings showed that the female adults exhibited higher scores on physiological symptoms ($M=6.40$, $SD=1.17$) as compared to the male adults ($M=5.28$, $SD=1.89$). The value of Cohen's d was .72 which indicates the large effect size. Further, there is a significant mean differences on remorse experiences of expressive reactions with $t(254) = 5.12$, $p < .001$. Findings showed that the females adults exhibited higher scores on expressive reactions ($M=6.08$, $SD=2.13$) as compared to the male adults ($M=4.75$, $SD=1.69$). The value of Cohen's d was .69 which indicates the large effect size. Correspondingly, there is a significant mean differences on remorse experiences of verbal reactions with $t(254) = 4.36$, $p < .001$. Findings showed that the female adults exhibited higher scores on expressive reactions ($M=4.06$, $SD=2.13$) as compared to the male adults ($M=3.05$, $SD=1.89$). The value of Cohen's d was .51 which indicates the medium effect size. Lastly, there is a significant mean differences on remorse experiences of psychological consequences with $t(254) = 2.83$, $p < .001$. Findings showed that the male adults exhibited higher scores on psychological consequences ($M=3.28$, $SD=1.31$) as compared to the female adults ($M=2.82$, $SD=1.58$). The value of Cohen's d was .68 which indicates the medium effect size. Hence, the hypothesis H_8 "there will be the difference in guilt, shame and remorse experiences between the male and female participants of Jammu & Kashmir and Uttar Pradesh" was supported in shame and remorse social emotion experiences but not in the context of guilt.